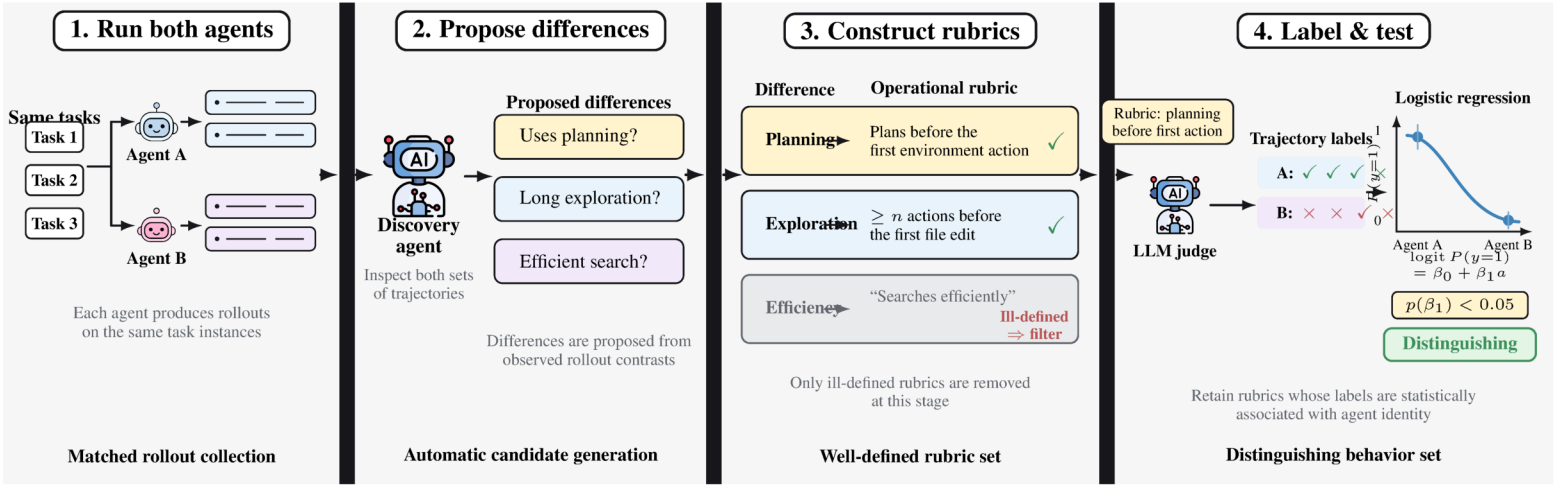

1. Run both agents
Same tasks
Task 1
Task 2
Task 3
Agent A
Agent B
Each agent produces rollouts on the same task instances
Matched rollout collection
2. Propose differences
AI
Discovery agent
Inspect both sets of trajectories
Proposed differences
Uses planning?
Long exploration?
Efficient search?
Differences are proposed from observed rollout contrasts
Automatic candidate generation
3. Construct rubrics
Difference
Operational rubric
Planning
Plans before the first environment action
Exploration
≥ n actions before the first file edit
Efficiency
"Searches efficiently"
Ill-defined ⇒ filter
Only ill-defined rubrics are removed at this stage
Well-defined rubric set
4. Label & test
Rubric: planning before first action
AI
LLM judge
Trajectory labels
A: ✓ ✓ ✓ ✗
B: ✗ ✗ ✓ ✗
Logistic regression
$\Pr(y=1)$
1
0
Agent A
Agent B
$\operatorname{logit} P(y=1) = \beta_0 + \beta_1 a$
$p(\beta_1) < 0.05$
Distinguishing
Retain rubrics whose labels are statistically associated with agent identity
Distinguishing behavior set